

Chapter 10

Continuously Compounded Interest

§ Section: 3.8

If an amount A_0 is invested at an annual interest rate r then after t years it's worth $A_0(1+r)^t$. However, interest is usually compounded more frequently, say, n times a year. Then in each compounding period the interest rate is r/n and there are nt compounding periods in t years.

	Interest rate for each period	Number of period in t years	After t years the investment is worth
Semiannual compounded	$\frac{r}{2}$	$2t$	$A_0 \left(1 + \frac{r}{2}\right)^{2t}$
Quarterly compounded	$\frac{r}{4}$	$4t$	$A_0 \left(1 + \frac{r}{4}\right)^{4t}$
Monthly compounded	$\frac{r}{12}$	$12t$	$A_0 \left(1 + \frac{r}{12}\right)^{12t}$
Daily Compounded	$\frac{r}{365}$	$365t$	$A_0 \left(1 + \frac{r}{365}\right)^{365t}$
Compounded n times a year	$\frac{r}{n}$	nt	$A_0 \left(1 + \frac{r}{n}\right)^{nt}$

$$t=5, r=0.02$$

$$A_0(1+0.01)^{10} = A_0 \cdot 1.1046$$

$$A_0 \left(1 + \frac{0.02}{4}\right)^{20} = A_0 \cdot 1.10489$$

$$A_0 \left(1 + \frac{0.02}{12}\right)^{60} = A_0 \cdot 1.10507$$

$$A_0 \left(1 + \frac{0.02}{365}\right)^{365 \times 5} = A_0 \cdot 1.10516$$

You can see that the interest paid increases as the number of compounding periods (n) increases. If we let n goes to infinity, then we will be compounding the interest continuously and the investment after t years will be worth

$$\lim_{n \rightarrow \infty} A_0 \left(1 + \frac{r}{n}\right)^{nt}.$$

The limit is $A(t) = A_0 e^{rt}$.

Example 1.1

Compute $\lim_{n \rightarrow \infty} A_0 \left(1 + \frac{r}{n}\right)^{nt}$.

Recall:

$$e = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$$

Sol: $\lim_{n \rightarrow \infty} \left(1 + \frac{r}{n}\right)^{nt}$

$$= \lim_{n \rightarrow \infty} \left(1 + \frac{1}{\frac{n}{r}}\right)^{\frac{n}{r} \cdot rt} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{x}\right)^{x \cdot rt} = e^{rt}.$$

Let $x = \frac{n}{r}$

Example 1.2

If 100000 NTD is invested at 2% interest, find the value of the investment at the end of 5 years if the interest is compounded (a) annually, (b) semiannually, (c) monthly, (d) continuously.

Sol: Annually: $10^5 \cdot (1 + 0.02)^5 =$

Semi: $10^5 \cdot \left(1 + \frac{0.02}{2}\right)^{10} =$

monthly: $10^5 \cdot \left(1 + \frac{0.02}{12}\right)^{60} =$

continuously: $10^5 \cdot e^{0.02 \times 5} = 10^5 \cdot e^{0.1}$

$A_0 \cdot e^{rt}$

Example 1.3

Assume that the interest rate is 3%, compounded continuously. How long will it take an investment to double in value?